All vertical class actions in the lower fifty-three tower

E.122. The stable positional fifty-three class tower and its actual unit-norm maps [Proved]

Keep the literal roots and fields of E.119 and E.121:

$$\begin{gathered}\zeta=\zeta_3,\quad \xi=\zeta_9,\quad \beta_d^{3^d}=53\zeta,\quad \beta_{d+1}^3=\beta_d,\\ B_d=\mathbb{Q}(\xi,\beta_d),\quad A_d=\mathrm{Cl}_S(B_d)\{3\},\quad S\mid\{3,53\}.\end{gathered}\tag{E122.1}$$

There are exactly three primes over three and three over fifty-three in every B_d, $d\ge1$. Each vertical step is totally ramified at all six of them and unramified elsewhere. For every $d\ge1$, the actual vertical automorphism $\sigma_d(\beta_{d+1})=\zeta\beta_{d+1}$, fixing ξ, acts identically on the entire group A_{d+1}. The actual ideal norms are onto and satisfy

$$N_d:A_{d+1}\twoheadrightarrow A_d,\qquad j_dN_d=3\,\mathrm{id}_{A_{d+1}},\qquad N_dj_d=3\,\mathrm{id}_{A_d}.\tag{E122.2}$$

For $d\ge2$ let $g_d(\beta_{d+1})=\xi\beta_{d+1}$, $g_d(\xi)=\xi$. This has order nine, fixes B_{d-1}, has cube σ_d, and restricts to σ_{d-1} on B_d. Its entire class action satisfies

$$(g_d-1)^2A_{d+1}=0,\qquad 3(g_d-1)A_{d+1}=0.\tag{E122.3}$$

Here are explicit local coordinates for every norm kernel in (E122.2). Order the six places by the three coherent branches at three, then the three fixed E.119 places at fifty-three. Put $U_d=\mathcal{O}_{B_d,S}^\times$ and let

$$\chi_d:U_d\longrightarrow\mathbb{F}_3^6,\qquad u\longmapsto\big(\log_\zeta(\mathrm{Art}_v(u)(\beta_{d+1})/\beta_{d+1})\big)_{v\in S}.\tag{E122.4}$$

The positive arithmetic Artin convention and the radical in this formula specify every sign. The actual local decomposition maps induce

$$\ker N_d\ \xleftarrow{\sim}\ \ker(\textstyle\sum:\mathbb{F}_3^6\to\mathbb{F}_3)/\chi_d(U_d).\tag{E122.5}$$

The restriction of χ_d to the included U_1 is exactly χ_1 and has rank two. Thus every kernel in (E122.2) is an elementary three-group of rank at most three. Equation (E122.5) retains the actual higher unit-character subgroup; it does not assign that subgroup the value of its included rank-two part. More precisely, put $K_d=\ker N_d$. The literal ideal extensions satisfy $K_d=j_dK_{d-1}$ for every $d\ge2$. Consequently $K_d\subset3A_{d+1}$, and every actual norm induces an isomorphism $A_{d+1}/3A_{d+1}\to A_d/3A_d$ for $d\ge2$. These four-dimensional vector spaces all have the dimension computed in E.121. The inverse class module and its exact filtration are described in (E122.20)–(E122.21). The explicit upper units in (E122.22) evaluate the first remaining kernel: $K_2=0$. Hence every K_d, $d\ge2$, is zero. Put $n_2=\mathrm{id}_{A_2}$ and, for $d>2$, $n_d=N_2\cdots N_{d-1}:A_d\to A_2$. These are actual isomorphisms and give

$$\begin{gathered}A_d\xrightarrow[n_d]{\sim}C_9\oplus C_3^3\quad(d\ge2),\qquad n_dN_d=n_{d+1},\qquad n_{d+1}j_d=3n_d,\\ \varprojlim_{d,N_d}A_d\xrightarrow[\mathrm{pr}_2]{\sim}A_2,\qquad j_d=3N_d^{-1}\quad(d\ge2).\end{gathered}\tag{E122.22a}$$

Here N_d^{-1} is the inverse of the proved isomorphism, and all coordinate maps and their compositions are specified. The actual g_d action is the identity for $d\ge2$. The horizontal action fixing β_d and sending ξ to ξ^4 is E.121's matrix H in the coordinates n_d.

The next longer cyclic extension B_3/B_1 is evaluated explicitly. Its generator is g_2, and its degree is nine. The maximal subfield of the S-Hilbert three-class field of B_3 that is abelian over B_1 has group

$$\Gamma_9=C_9^2\oplus C_3^3,\qquad \ker\{\Gamma_9\longrightarrow\langle g_2\rangle\}=(A_3)_{g_2}=C_9\oplus C_3^3.\tag{E122.6}$$

The complete nine-coordinate relation lattice, its Smith maps and inverse maps are given in the certificate. In their four nontrivial Smith coordinates, the literal norm to the full group in E.121 is

$$\begin{aligned} \overline{N}_2 : (A_3)_{g_2} \xrightarrow{\sim} A_2, \qquad (x_0, x_1, x_2, x_3) \longmapsto (x_0, x_1, x_2, x_3), \\ \bar{j}_2 = 3I_4, \qquad \ker N_2 = (g_2 - 1)A_3. \end{aligned} \tag{E122.7}$$

The first coordinate has modulus nine and the other three have modulus three. The horizontal automorphism fixing β_3 and sending ξ to ξ^4 acts on this quotient by exactly E.121's matrix H, transported by the displayed norm isomorphism. The subgroup on the right of (E122.7) vanishes by the literal unit witnesses below. The computation of this quotient is retained with its actual comparison map as an independent finite check. These are literal root-cubing and ideal-norm maps for the $j = 1$ positional factor of E.46. The full embedded first unit-norm subgroup is evaluated by (E122.23), and every later finite unit-norm defect and completed kernel is evaluated by (E122.25). The actual higher Hilbert90 maps specify their embedded subgroups. The maps from these positional norm systems to the intrinsic section groups remain the separate maps of E.46 and E.23. The manuscript anchor is §51.16, fols. 733v–736v, scans172–178, through the exact root and Kummer–Shapiro comparisons of E.46.

Proof. Six coherent places and their actual uniformizers. Let a_0 be the rational three-adic cube root of fifty-three congruent to -1 modulo nine. Since $53 = -1 + 54$, the logarithm on $1 + 3\mathbb{Z}_3$ gives $v_3(a_0 + 1) = 2$. The three embeddings of B_1 over $\mathbb{Q}_3(\xi)$ send $\alpha = 53^{1/3}$ to $a_0\zeta^j$, $j = 0, 1, 2$, and hence send β_1 to $a_0\xi^{1+3j}$. Normalize the valuation of $F_3 = \mathbb{Q}_3(\xi)$ by $v(1 - \xi) = 1$, so $v(3) = 6$. For $k = 1, 4, 7$,

$$a_0\xi^k + 1 = (1 - \xi^k) + (a_0 + 1)\xi^k, \qquad v(1 - \xi^k) = 1, \quad v(a_0 + 1) = 12. \tag{E122.8}$$

Thus $\beta_1 + 1$ is a uniformizer at each labeled place. Whenever $\beta_d + 1$ is a uniformizer, the polynomial

$$(Y - 1)^3 - \beta_d = Y^3 - 3Y^2 + 3Y - (\beta_d + 1) \tag{E122.9}$$

is Eisenstein there. It defines the unique next prime, with unchanged residue field, and $\beta_{d+1}+1$ is its uniformizer. At each fifty-three prime, β_1 has valuation one and $X^3 - \beta_d$ gives the same Eisenstein induction, with uniformizer β_{d+1}. The initial number and residue degrees of these places are E.119's literal splitting calculation. The defining polynomial discriminants have support only at three and fifty-three. This proves all the asserted local degrees, uniqueness of primes in each vertical chain, and the absence of other ramification.

The class-field kernel with its decomposition generators. Write $D = B_d$, $E = B_{d+1}$ and let H_E^S be the maximal abelian three-extension of E unramified everywhere and split at every place of S. It is invariant under every automorphism of E preserving S. Put $\mathcal{G} = \mathrm{Gal}(H_E^S/D)$ and $\Gamma = \mathcal{G}^{\mathrm{ab}}$. The actual ideal Artin maps identify A_E with $\mathrm{Gal}(H_E^S/E)$, and give

$$0 \longrightarrow A_E/(\sigma_d - 1)A_E \longrightarrow \Gamma \longrightarrow C_3 \longrightarrow 0. \tag{E122.10}$$

Indeed A_E is abelian, the quotient is cyclic, and the commutator subgroup is exactly $(\sigma_d-1)A_E$. A decomposition group at any S-place gives a section of the last map: its restriction to E/D is an isomorphism, because E/D is totally ramified and the place splits in H_E^S/E.

There are two class-field inclusions to retain. Base change gives $H_D^S E \subset H_E^S$, with the prescribed splitting at S. Moreover $E \cap H_D^S = D$, since a nontrivial subextension of the cyclic cubic E/D is ramified at S. Restriction consequently gives the actual onto ideal norm $A_E \to A_D$. The largest subfield of H_E^S that is abelian over D, unramified everywhere and split at S is exactly H_D^S. Thus

$$0 \longrightarrow I \longrightarrow \Gamma \longrightarrow A_D \longrightarrow 0, \qquad I = \langle \text{the six S-decomposition images} \rangle. \tag{E122.11}$$

These statements use the literal Frobenius norm square and the trivial-modulus class-field correspondence [1, V,Proposition3.3, Theorem3.6 and Example3.9]. The kernel description follows by killing these decomposition groups: outside S the field is already unramified, and killing the S-groups makes every S-prime split. Conversely H_D^S is contained by the first inclusion. Each generating group is cyclic of order three and injects into $\mathrm{Gal}(E/D)$.

The all-level normalizer induction. The base case $A_2^{\sigma_1} = A_2$ is E.121's full class theorem. For $d \geq 2$, use the actual $g = g_d$ in the statement. Its restriction to D is the preceding vertical automorphism, so it acts identically on A_D by induction. It fixes each S-prime of D, since the preceding vertical step has a unique prime above each one. It also commutes with σ_d. Consequently its action on every decomposition image in (E122.11) is the identity: restriction embeds that image into $\langle \sigma_d \rangle$, on which conjugation by g is the identity. Choice of a lift of g to H_E^S changes its action on Γ by an inner automorphism, hence does not change it. For $T = g - 1$ these actual comparisons give

$$T\Gamma \subset I, \qquad TI = 0, \qquad T^2\Gamma = 0. \tag{E122.12}$$

On the entire still unknown finite group $A = A_E$, put $s = g^3 - 1$ and $M = sA$. From (E122.10)–(E122.12) one has $T^2 A \subset M$. Actual norm of an ideal followed by extension is the product of its conjugates, so $jN = 1 + \sigma_d + \sigma_d^2 = 3 + 3s + s^2$ on A. Because g acts trivially on A_D, $TjN = 0$. It follows that

$$3TA = -T(3+s)M, \qquad M = (3T + 3T^2 + T^3)A \subset (3, T)M. \tag{E122.13}$$

For $3^e A = 0$, the exact relation $(1+T)^9 = 1$ gives $T^9 \in 3\mathbb{Z}_3[g]$. Every monomial $3^a T^b$ of total degree $9e$ is therefore divisible by at least $3^{a+\lfloor b/9 \rfloor}$, and $a + \lfloor b/9 \rfloor \geq e$. Hence $(3, T)^{9e} A = 0$. Iteration of (E122.13) proves $M = 0$. This proves the entire vertical action, not only its coinvariants. Now (E122.12) gives $T^2 A = 0$, and $g^3 = 1$ on A gives $3T = 0$. The ideal identities yield (E122.2).

Every remaining kernel as an actual unit-character quotient. The product of local reciprocity maps identifies the subgroup I in (E122.11) with $\mathbb{F}_3^6/\chi_d(U_d)$: the local norm subgroup is killed at S, local units are killed outside S, and the kernel among S-ideles is precisely the diagonal S-unit image. This is the idele sequence with its explicit local maps [1, V,Proposition5.2 and Theorems5.3 and5.5]. Restriction to E/D sends each positive local generator to σ_d, and so is the sum of the six coordinates. The global product formula puts $\chi_d(U_d)$ in its kernel. Intersecting (E122.10) and (E122.11), now with $sA = 0$, proves (E122.5) and its stated inverse: form the corresponding product of actual local Artin elements and read its class in A_E.

For $u \in U_1$ the local corestriction projection formula gives

$$\mathrm{Cor}\big([\beta_d] \cup \mathrm{Res}[u]\big) = [N_{B_{d,v}/B_{1,v_1}} \beta_d] \cup [u] = [\beta_1] \cup [u]. \tag{E122.14}$$

The norm is exactly β_1, from the odd-degree binomial minimal polynomial and the Eisenstein local degree. The local invariant composed with corestriction is the lower invariant, so there is no factor of the local degree. This is the proved cup-product Res/Cor compatibility of [1, III,Theorem3.1 and its proof]; it also follows after taking a finite normal closure when the longer local extension is not Galois. The fixed μ_3 is transported by its literal inclusion. Thus (E122.14) is exactly $\chi_d(u) = \chi_1(u)$. E.119's complete rank-two matrix proves the bound in (E122.5).

Complete local ninth-norm characters at the next stage. Here B_3/B_1 is cyclic of degree nine, generated by g_2. At a labeled three-prime it is $F_3(\theta)/F_3$ with $\theta^9 = a_0 \xi^{1+3j}$; $\theta + 1$ is an Eisenstein uniformizer. Let $\lambda = 1 - \xi$. The logarithm and exponential show $1 + \lambda^{16}\mathcal{O}_{F_3} \subset (F_3^\times)^9$: division by nine lowers the λ-valuation by twelve and leaves valuation at least four, in their

inverse range. The finite unit quotient modulo ninth powers therefore has seven cyclic factors of order nine. The exact residue-ring generators used by the program, in order, are

$$\begin{gathered} 1+\xi, \quad -\xi-3\xi^2-3\xi^3-\xi^4, \quad 3+\xi^3, \\ -1+2\xi^2+2\xi^3+\xi^5, \quad -2\xi^2-2\xi^3-\xi^5, \quad 1+3\xi, \quad 2+2\xi+2\xi^3+\xi^4. \end{gathered} \tag{E122.15}$$

Their exact residue-group orders and discrete-log inverse are computed in $\mathcal{O}_{\mathbb{Q}(\xi)}/\lambda^{16}$, with the official `idealstar`/`ideallog` maps [2, those function contracts].

For each of the fifty-five indices $1 \le i \le 80$, $3 \nmid i$, and $i = 81$, compute the literal norm

$$n_{j,i} = \operatorname{Res}_X\big((X-1)^9 - a_0\xi^{1+3j}, 1+X^i\big). \tag{E122.16}$$

These are norms of actual local units. The certificate proves that their images generate a subgroup of index exactly nine in the finite unit quotient. The local norm-index theorem proves that the complete norm subgroup also has index nine, since this is a totally ramified cyclic extension of degree nine. Inclusion and the identical index prove equality; no completeness assertion about the fifty-five upper units is required. In the coordinates (E122.15) the three positive character rows are

$$\begin{pmatrix} 3 & 6 & 7 & 3 & 0 & 3 & 0 \\ 3 & 6 & 1 & 3 & 0 & 3 & 0 \\ 3 & 6 & 4 & 3 & 0 & 3 & 0 \end{pmatrix} \mod 9. \tag{E122.17}$$

The sign and generator are calibrated by the actual unit four. For the rational factors $a_0, 4$, naturality under complex conjugation fixes both arguments and inverts μ_9, so their odd-order Hilbert symbol is one. On the radical of ξ, arithmetic local reciprocity acts on ζ_{81} by the inverse of $N_{F_3/\mathbb{Q}_3}(4) = 4^6$. Since $4^{-6} = 37 \pmod{81}$, the resulting exponent is four modulo nine. Thus the values on four are $4(1+3j)$, namely $4, 7, 1$. These unit values specify each row uniquely.

The program uses an integer approximation to a_0 whose cubing error has valuation at least twenty-four. Its ratio to the exact a_0 is a ninth power by the same logarithmic calculation. Multiplying θ by that literal local ninth root identifies the approximate and exact Kummer fields and preserves the chosen generator. When evaluating a lower global element, the recorded coefficient and divisor bounds prove that its approximate normalized local unit differs from the exact one by $1+\lambda^{109}\mathcal{O}$ or better. Thus all displayed characters are exact, not inferred from agreement of two finite precisions.

At fifty-three normalize an element by its valuation power of β_1. Its positive character is the negative logarithm to the residue of ξ of the normalized residue raised to $(53^2-1)/9$. The full six-by-fifteen matrix on the actual complete E.119 S-unit basis, each class-power principal correction, and each S-prime principal correction is retained. Every column satisfies the ninth product formula and reduces to the exact old cubic column. The three outside-S class generators have ninth Frobenius coordinates $(2, 1, 6)$. Consequently the field restriction row on the nine-coordinate genus presentation is

$$(1,1,1,1,1,1,2,1,6) \pmod 9. \tag{E122.18}$$

The relation HNF, with all forward and inverse Smith maps, is computed from these actual principal ideals. Its full group has orders $(9,9,3,3,3)$ and its restriction kernel has orders $(9,3,3,3)$. The kernel is $(A_3)_{g_2}$ by the same actual class-field argument as (E122.10), now of degree nine. This proves (E122.6).

Literal reduction to the preceding genus field. The inclusion $H^S_{B_2}B_3 \subset H^S_{B_3}$ is the actual base change above. E.121 proves that $H^S_{B_2}/B_1$ is already abelian. On the six local coordinates,

restricting the ninth radical to its cube is reduction modulo three; on the three fixed ideal Frobenius coordinates it retains the same ideal. The ambient nine-coordinate map is therefore the identity. The certificate checks inclusion of the complete new relation lattice in the old one, then transports this map through both restriction-kernel lattices and Smith bases. Its four nontrivial coordinates are literally I_4. All 243 elements and both finite inverse compositions are independently checked. Naturality gives the stated horizontal matrix, and the actual trace $j_2 N_2 = 3$ gives $\overline{j}_2 = 3I_4$. Since the induced norm on g_2-coinvariants is injective, its full kernel is exactly $(g_2 - 1)A_3$, proving (E122.7).

The actual recursion of norm kernels. Fix $d \geq 2$, $E = B_{d+1}$, $D = B_d$, and $D_0 = B_{d-1}$. The full field H_E^S is Galois over D_0, because the cyclic-nine extension E/D_0 preserves S. At each of its six S-places choose the unique element γ_i of the actual local decomposition group that restricts to g_d on E. That decomposition group is cyclic of order nine, since the place splits over E. Fix one as γ_0 and write $\gamma_i = \gamma_0 a_i$ with $a_i \in A_E$. Conjugation by γ_0 is the actual g_d action. By (E122.3), $1 + g_d + g_d^2 = 3$ on A_E. The group law therefore gives

$$\gamma_i^3 \gamma_0^{-3} = 3a_i = j_d N_d(a_i), \qquad K_d = j_d K_{d-1} \subset 3A_{d+1}. \tag{E122.19}$$

Here is the exact restriction comparison giving the second equality. The cubes γ_i^3 are the positive S-decomposition generators in $\mathrm{Gal}(H_E^S/D)$; their differences from the reference generate K_d by (E122.5). Restriction through $H_D^S \subset H_E^S$ sends γ_i to the positive lower decomposition generator in $\mathrm{Gal}(H_D^S/D_0)$. Their differences are $N_d(a_i)$ by the ideal Frobenius norm square, and generate K_{d-1}. Thus (E122.19) is equality of actual subgroups, not only an order comparison. Changing a prolongation conjugates γ_i by an element of A_E and changes a_i by a value of $g_d - 1$; its triple is zero by (E122.3). Changing the reference replaces all differences by differences from another generator and leaves both generated subgroups unchanged. This records the choice ambiguity in the literal local maps. The inclusion in (E122.19) and surjectivity of N_d imply the stated mod-three isomorphisms. The ranks of the elementary groups K_d are nonincreasing from the rank-three group K_1 of E.121.

An exact finite-rank inverse class module. Let $\mathcal{A} = \varprojlim_{d,N_d} A_d$, and put $\mathcal{B}_d = \ker(\mathcal{A} \to A_d)$. Every projection is onto, by compactness and the onto finite norms. The actual kernel recursion gives

$$\mathcal{B}_{d+1} = 3\mathcal{B}_d \quad (d \geq 2), \qquad A_d = \mathcal{A}/3^{d-2}\mathcal{B}_2 \quad (d \geq 2), \qquad \mathcal{B}_2/3\mathcal{B}_2 = K_2. \tag{E122.20}$$

To check all limit steps, for $e > d$ put $M_{e,d} = \ker(A_e \to A_d)$. The restricted top norm maps $M_{e,d}$ onto $M_{e-1,d}$. Thus $3M_{e,d} = j_{e-1}M_{e-1,d}$, by the actual trace identity. Induction proves $3M_{e,d} = M_{e,d+1}$: its top norm image is $M_{e-1,d+1}$ by the preceding induction step, and it contains the entire top kernel $j_{e-1}K_{e-2} = K_{e-1}$ by (E122.19). The initial case $e = d + 1$ has both sides zero. For any prescribed element of $\mathcal{B}_{d+1}$, the finite sets of solutions of $3x = y$ in these compatible finite kernels give nonempty closed subsets of the compact group $\mathcal{B}_d$. Any finite intersection is nonempty by working at the largest level. Compactness proves (E122.20) with no omitted inverse-limit term.

The group $\mathcal{B}_2$ is generated as a compact $\mathbb{Z}_3$-module by at most three elements. Explicitly choose lifts of a basis of $\mathcal{B}_2/3\mathcal{B}_2 = K_2$; successive reduction modulo $3^n\mathcal{B}_2$ writes every element as a convergent $\mathbb{Z}_3$-linear combination of those lifts. For each finite quotient the remainder tends to zero, since that quotient has finite three-power exponent. This is the required topological Nakayama argument. Since $\mathcal{B}_2$ has finite index in $\mathcal{A}$, the actual limit satisfies

$$\mathcal{A} \text{ is a finitely generated } \mathbb{Z}_3\text{-module}, \quad \mathrm{rank}_{\mathbb{Z}_3}\mathcal{A} \leq 3, \quad \mathcal{A}/3\mathcal{A} \simeq A_2/3A_2 \simeq \mathbb{F}_3^4. \tag{E122.21}$$

The last map is the actual projection: inverse limits of the finite multiplication-by-three sequences are exact, and their mod-three transition maps are the isomorphisms already proved.

Neither the rank nor the torsion exponents of this embedded limit module are assigned by (E122.21). Equation (E122.20) retains its literal finite-index submodule, rather than replacing it by an unspecified isomorphism type.

Literal upper units that evaluate the remaining kernel. The compact witness file uses the exact relative field $\mathbb{Q}[b,z]/(z^6+z^3+1, b^9-53z^3)$, with $b=\beta_2$, $z=\xi$ and lower primitive element $z-b^3/z$. The relative binomial is irreducible by the fifty-three Eisenstein valuation. The file gives all rational coefficients in the ordered basis $b^i z^j$, $0\le i<9$, $0\le j<6$. It also gives ten literal elements of $\mathbb{Q}(b)$ and twenty-two elements $v_1,\dots,v_{22}$ of B_2. Each of the former has its entire principal divisor verified to have support only at three and fifty-three. Include the fifteen certified lower S-units and the three g_1-conjugates of each of these ten elements, giving forty-five actual S-units w_i of B_2. Here g_1 denotes the literal automorphism $b\mapsto \xi b$, $\xi\mapsto\xi$ of $B_2/\mathbb{Q}(\xi)$; its cube is σ_1. For every v_k the certificate verifies the literal identity $v_k^3=\prod_{i=1}^{45} w_i^{a_{ki}}$, with every integer a_{ki} displayed. In particular every v_k is an S-unit: outside S its valuation is one third of a zero valuation. A separate rational relative-ring program rechecks all twenty-two identities by cross-multiplication, without a number-field library or a root search.

The same two programs verify every relative norm $v_k\sigma_1(v_k)\sigma_1^2(v_k)$, its exact expression in the complete lower S-unit basis, and its sign. The actual local norm square sends the action on β_3 over B_2 to its action over B_1. With $\zeta=\xi^3$, it therefore gives exactly $\chi_9(N_1v_k)=3\chi_2(v_k)$, with the threefold map $\mathbb{F}_3\to\mathbb{Z}/9$ specified by these roots. Applying the already calibrated rows (E122.17) and the exact tame residue formula proves all six characters of every witness. The minimum proved approximation bound is $1+\lambda^{111}\mathcal{O}$. For lower units u_8,u_{10} in the fixed fifteen-column order and the three roots v_8,v_{14},v_{20} the actual character matrix is

$$\bigl(\chi_2(u_8),\chi_2(u_{10}),\chi_2(v_8),\chi_2(v_{14}),\chi_2(v_{20})\bigr)=\begin{pmatrix}0&1&2&0&2\\0&1&0&1&2\\0&1&1&2&1\\2&0&0&0&2\\2&0&1&0&1\\2&0&2&0&1\end{pmatrix}\pmod 3. \tag{E122.22}$$

Every column has sum zero and the first five rows have determinant -4. Thus $\chi_2(U_2)$ is the entire product-formula hyperplane. Equation (E122.5) proves $K_2=0$, and (E122.19) proves $K_d=0$ for every $d\ge 2$. This proves all isomorphisms (E122.22a), including their displayed inverses and compositions. Norm equivariance shows that g_d acts identically and that the horizontal matrix is H at every level. It also evaluates the earlier inverse filtration: $\mathcal{B}_2=0$ and $\mathcal{A}=A_2$ through the actual projection. The preliminary finite-prime residue rank and the smaller-field candidate lattice are retained as discovery records. Only the literal cube, divisor, norm and positive-character identities enter this conclusion; no completeness claim about an upper unit basis or an upper BNF is used.

The full embedded first unit-norm subgroup and an inverse. The actual fifteen-column basis identifies $\widehat{U}_1$ with $\mathbb{Z}_3^{15}/\mathbb{Z}_3(9,0,\dots,0)$, the first column being the literal root ξ. The index-nine inverse-image lattice of $\ker\chi_1$ consists exactly of the lower S-units that are field norms from B_2, by the cyclic Hasse norm theorem. E.121's literal Hilbert90 divisor map identifies the quotient of those field-norm units by actual S-unit norms with $A_2/j_1A_1=C_3^4$. Thus the complete unit-norm subgroup has index 3^6 in $\widehat{U}_1$.

Write $c_k\in\mathbb{Z}^{15}$ for the displayed norm-coordinate column of v_k, and $s_k\in\{1,-1\}$ for its displayed sign, so $N_1v_k=s_k\prod u_i^{(c_k)_i}$. The group generated by these columns and $3I_{15}$ has exact HNF determinant 729. All columns are actual norms, since $N_1(s_kv_k)=\prod u_i^{(c_k)_i}$ and $N_1(u_i)=u_i^3$ for the included lower unit. Equality of the proved indices shows that this is the

entire completed norm subgroup. Its literal congruence description is

$$N_1\widehat{U}_2 = \{[x] \in \mathbb{Z}_3^{15}/\mathbb{Z}_3(9e_1) : Qx = 0 \pmod 3\},$$
$$Q = \begin{pmatrix} 0&1&0&0&2&0&1&0&1&0&2&0&0&0&0 \\ 0&0&1&0&1&0&2&0&1&0&2&0&2&1&0 \\ 0&0&0&1&2&2&1&0&0&0&1&0&0&0&0 \\ 0&0&0&0&0&0&0&1&2&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&1&1&0&1&1&1 \\ 0&0&0&0&0&0&0&0&0&0&0&1&0&0&0 \end{pmatrix}. \tag{E122.23}$$

The six coordinate columns numbered $2, 3, 4, 8, 10, 12$ give a right inverse to Q modulo three. For an actual preimage map on the norm image, let M have first nine columns $c_1, c_8, c_9, c_{10}, c_{11}, c_{12}, c_{14}, c_{15}, c_{20}$ and last six columns $3e_2, 3e_3, 3e_4, 3e_8, 3e_{10}, 3e_{12}$. Its determinant is $94770 = 3^6 \cdot 130$. The certificate gives the complete rational inverse and checks that its lattice over $\mathbb{Z}_3$ is exactly (E122.23), in both directions. For a representative x in that lattice, form $a = M^{-1}x \in \mathbb{Z}_3^{15}$ and take the corresponding product of the nine actual units $s_k v_k$ and the six included units $u_2, u_3, u_4, u_8, u_{10}, u_{12}$, with exponents a. Its norm is exactly $\prod u_i^{x_i}$. Changing x by $9e_1$ changes this preimage by an actual norm-one element. Every other preimage is its product with the completed norm kernel evaluated in E.121. This specifies the entire fibre and its torsion-coordinate ambiguity, rather than only an index comparison.

All later unit defects and their completed norm kernels. For every $d \geq 2$, the local corestriction comparison (E122.14), now starting at B_2, transports the five columns (E122.22) unchanged to χ_d. Hence its image has rank five. Set $F_d = U_d \cap N_d(B_{d+1}^\times)$. The literal maps are

$$\begin{aligned} &U_d/F_d \xrightarrow[\chi_d]{\sim} \ker \Sigma \subset \mathbb{F}_3^6, \qquad F_d/N_d U_{d+1} \xrightarrow[\delta_d]{\sim} A_{d+1}/j_d A_d \xrightarrow[n_{d+1}]{\sim} A_2/3A_2, \\ &\delta_d(u) = [D_x] \pmod{j_d A_d}, \qquad N_d x = u, \qquad \operatorname{div}_{S^c}(x) = (\sigma_d - 1)D_x. \end{aligned} \tag{E122.24}$$

In the divisor class take the canonical three-primary projection. The precise Hilbert90 construction, its well-definedness under a different x and D, and both kernel and surjectivity are the ideal maps proved in E.121. Here all vertical class actions are trivial, so its image is the entire displayed quotient. The actual class transport n_{d+1} evaluates that quotient as C_3^4. These two maps describe the higher embedded norm subgroup exactly: first $\chi_d(u) = 0$, then $\delta_d(u) = 0$. Their completions retain the same finite quotients.

Every lower cube is the norm of the included element, so the full cokernel is killed by three. Its order is $3^5 \cdot 3^4 = 3^9$. For the torsion in the completed kernel, put $F = \mathbb{Q}(\xi)$. At each fifty-three place of F, $X^{3^d} - 53\zeta$ is Eisenstein, so B_d/F has a unique totally ramified prime there. The extension $F(\mu_{27})/F$ splits completely at that place, because $27 \mid 53^2 - 1$. Its inclusion in B_d would, after tensoring with that completion of F, inject a product of three fields into the single field $B_{d,v}$, which is impossible. Thus $\mu(B_d)\{3\} = \mu_9$ at every level. The actual norm on this subgroup is cubing and has kernel the literal μ_3. Dirichlet's S-unit rank is $3^{d+1} + 5$, since $[B_d : \mathbb{Q}] = 6 \cdot 3^d$ and S has six places. The norm image has finite index, so

$$\widehat{U}_d/N_d\widehat{U}_{d+1} = C_3^9, \qquad \ker(N_d : \widehat{U}_{d+1} \to \widehat{U}_d) \simeq \mu_3 \oplus \mathbb{Z}_3^{2\cdot 3^{d+1}} \quad (d \geq 2). \tag{E122.25}$$

The last isomorphism has an actual construction: choose a $\mathbb{Z}_3$-basis of the kernel modulo its literal torsion and choose lifts k_i; send $(z, (a_i))$ to $z \prod k_i^{a_i}$. The inverse reads quotient coordinates and then divides by that product to obtain z. It is a continuous bijection between compact groups, hence a homeomorphism. Changing the basis or its lifts changes these coordinates by the corresponding invertible integral matrix and by the stated μ_3 factors. No numerical upper unit basis is assigned by this choice. Equations (E122.23)–(E122.25) preserve the actual norm subgroups and the exact class and divisor maps required for the positional Kummer–Shapiro inverse systems. □

References

[1] J. S. Milne, *Class Field Theory*, version4.03, 2020.

[2] The PARI Group, *PARI/GP2.19.0 User's Guide*, official function contracts and implementation.